MOEEZ AHMED



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OBJECTIVES

To pursue a master's degree in China and grow into an independent researcher in machine learning, focusing on reliable recognition, rigorous evaluation and calibration, and the trustworthy use of large language models. I want to learn research methods under expert supervision.

ACADEMIC QUALIFICATION

BS Computer Science (3.72 / 4.00 CGPA) *2022 – 2026*
COMSATS University Islamabad, Lahore Campus, Pakistan

Specialized in machine learning, artificial intelligence, algorithms, information security and software engineering. Course grades (out of 4.00): Machine Learning 4.00, Linear Algebra 4.00, Algorithms 4.00, Information Security 4.00, Artificial Intelligence 3.66. Final Year Project (6 credit hours): 4.00, supervised by Prof. Dr. Zulfiqar Habib.

Intermediate, Pre-Engineering (Grade A) *2020 – 2022*
Government College University (GCU), Lahore, Pakistan

Matriculation (Grade A) *2018 – 2020*
Beaconhouse School System, Pakistan

RESEARCH DIRECTION

- Machine Learning and Deep Learning
- Metric Learning and Open-Set Recognition
- Model Evaluation and Calibration
- Large Language Models and Retrieval-Augmented Generation
- Hallucination and Faithfulness in LLMs
- Multimodal Learning

RESEARCH AND PROJECTS

Multi-Camera Face Recognition Attendance System *Industry project, 2026*

- **Built:** an end-to-end pipeline (YuNet detection, 5-point alignment, SFace embeddings, cosine matching) on live multi-camera streams.
- **Found:** field misidentification at distance showed the threshold needed recalibration (0.36, then 0.51, then 0.46). I analysed the false-accept versus false-reject trade-off through inter-class and intra-class similarity margins, and compared the lightweight-versus-accuracy trade-off of SFace and ArcFace-class models.
- **Evidence:** on LFW (6,000 pairs, 10-fold), the pretrained models reach **99.1% ± 0.4% accuracy, 1.05% EER and AUC 0.9965**. The deployed threshold gave **0 false accepts in 3,000 impostor pairs** at a 5.5% false-reject rate.

TABEEB: LLM-Based Multimodal Report Analysis Platform *Final Year Project*

- **Built:** an LLM-based multimodal pipeline that accepts scanned images, PDFs and text and returns patient-readable summaries flagging values outside reference ranges.
- **Studied:** prompt design and decoding control in a high-stakes domain (low temperature for factual summaries, higher for conversational Q&A) to locate where factual fidelity breaks down. Context summarization cut token usage by about 30%.
- **Designed:** a retrieval-augmented pipeline where the model formats only retrieved facts, with labelled estimates as a fallback, and made the system work across model families (Gemini and Gemma), which showed how instruction following varies between models.
- **Open problems:** faithfulness of LLM summarization, evaluating output without ground truth, low-resource Urdu-English NLP.

WORK EXPERIENCE

Software Engineer, Inspirer Technologies, Lahore *Jun 2026 – Present*

- Independently designed, built and field-tuned a multi-camera face-recognition attendance system (see Research and Projects), recalibrating its matching threshold after observing distance-induced misidentification.
- Own the HR and Finance modules of a company-wide ERP, translating payroll and financial rules into a consistent relational data model for employee, payroll and financial records.

Backend Developer Intern, Amzone, Lahore *Jul – Aug 2025*

- Investigated and fixed data-synchronization problems in the wish-list feature of a production e-commerce platform, and designed the database structure for its product, order and inventory data.

ACADEMIC PROJECTS

- **TABEEB**, LLM-based multimodal report analysis platform (Final Year Project).
- **Traffic signal detection**, semester project of the Computer Vision course, developed in MATLAB.
- **Peer-to-peer file sharing service**, with a custom mechanism for peer discovery and transfer coordination.

HONORS & ACHIEVEMENTS

- **Runner-up, Projects Category, VisioSpark competition** (for Tabeeb, the final year project).
- **4.00 / 4.00 GPA in Semester 7**, including the 6-credit-hour Final Year Project (4.00 / 4.00).
- **4.00 / 4.00 in Machine Learning, Information Security, Linear Algebra and Algorithms.**

TRAININGS & SKILLS

Programming: Python, MATLAB, SQL
Machine Learning: PyTorch, scikit-learn, OpenCV (YuNet, SFace), embedding-based recognition, ROC / EER / TAR-at-FAR evaluation, threshold calibration
LLMs: LLM APIs (Gemini), prompt design, decoding control, retrieval-augmented generation
Tools: Git, GitHub, Docker

EXTRA-CURRICULAR ACTIVITIES

- Played football and basketball for my academic institutions.
- Participated in several AMD hackathons (team-based problem solving under time limits).
- Mentored junior students in several core computer science subjects.
- Volunteered at blood donation drives.

PERSONAL DATA

Nationality: Pakistani
Address: Lahore, Punjab, Pakistan
Languages: Urdu (native); English (degree taught in English; medium-of-instruction certificate available)

REFERENCES

(1) Prof. Dr. Zulfiqar Habib
Former Dean and Tenured Professor, Department of Computer Science, COMSATS University Islamabad, Lahore Campus (Final Year Project Supervisor)
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(2) Dr. M. Hasan Jamal
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